

Frank Hu

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RESEARCH INTERESTS

Generative AI and foundation models (transformers, large language models, chemical language models, vision language models, diffusion models, flow matching models) for molecule and protein design; gradient-based post-training algorithms for molecular optimization and inverse design problems; physics-informed neural networks for molecular and biomolecular simulations and modeling.

EDUCATION

Stanford University

PhD Candidate in Chemistry, GPA: 4.00/4.00

Stanford, CA

August 2022 - Present

Carnegie Mellon University

MS in Chemistry, GPA: 3.97/4.00

BS in Chemistry, GPA: 3.97/4.00

Pittsburgh, PA

August 2020 - August 2022

August 2018 - August 2022

RESEARCH

Prescient Design/AI4DD, Genentech

Machine learning research intern with the foundation models team

South San Francisco, CA

June 2026 - Present

Markland Lab and Rotskoff Lab, Department of Chemistry, Stanford University

Graduate researcher with Professor Thomas E. Markland and Professor Grant M. Rotskoff

Stanford, CA

August 2022 - Present

- Building foundational chemical language models and machine learning frameworks for small molecule design, optimization, and conformational sampling
- Building chemical language model-based frameworks for direct interpretation of spectroscopic data
- Developing data-efficient machine-learned force fields for condensed phase molecular dynamics

Yaron Lab, Department of Chemistry, Carnegie Mellon University

Undergraduate and Masters researcher with Professor David J. Yaron

Pittsburgh, PA

August 2018 - August 2023

- Built a novel framework for machine-learned parameterization of semiempirical chemical Hamiltonians
- Thesis title: *DFTBML: A Machine-Learned Density Functional Based Tight Binding Model for the Deep Learning of Chemical Hamiltonians*

Jiang Lab, Department of Mathematics and Statistics, Georgia State University

Research intern with Professor Yi Jiang

Atlanta, GA

June 2017 - April 2018

- Developed computational methods for analyzing cancer cell metastasis based on microscopy image data

PUBLICATIONS AND PRESENTATIONS

Publications

- 1 N. I. Hausman, J. Kelly, M. S. Chen, **F. Hu**, A. lee, A. Montoya-Castillo, G. S. Schlau-Cohen, T. E. Markland. Streamlining Analysis and Design of Two-Dimensional Electronic Spectroscopy using Machine Learning. *Under review* (2026)
- 2 J. Liu, S. Ibarraran*, **F. Hu***, A. Park*, A. Dunn, G. M. Rotskoff. Generative design of intrinsically disordered protein protein regions with IDiom. *ICML 2026 workshop on Generative and Agentic AI for Biology* (2026)
- 3 J. K. Rasch, A. L. Clayborn, D. Sole-Barber, S. Wan, C. A. Saavedra, A. J. Fairhall, **F. Hu**, J. Bartz, Y. Podorova, A. J. Bergsten, Z. D. Young, I. D. R. Barrera, T. H. K. Mone, N. Asgar, T. E. Markland, R. H. Goldsmith. Label-Free Optical Differentiation of Single Diffusing Amino Acids at Picomolar Concentrations. *Under review* (2026)
- 4 S. Ibarraran, S. Chennakesavalu, **F. Hu**, G. M. Rotskoff. Efficient, Few-shot Directed Evolution with Energy Rank Alignment. *ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design* (2026)
- 5 **F. Hu**, J. M. Tubb, D. Argyropoulos, S. Golotvin, M. Elyashberg, G. M. Rotskoff, M. W. Kanan, T. E. Markland. Pushing the limits of one-dimensional NMR spectroscopy for automated structure elucidation using artificial intelligence. *In press, JCIM* (2026)
- 6 S. Chennakesavalu, **F. Hu**, S. Ibarraran, and G. M. Rotskoff. Aligning Transformers with Continuous Feedback via Energy Rank Alignment. *The Thirty-ninth Annual Conference on Neural Information Processing Systems* (2025)

- 7 J. Kelly, F. **Hu**, A. Damiani, M. S. Chen, A. Snider, M. Son, A. Lee, P. Gupta, A. Montoya-Castillo, T. J. Zuehlsdorff, G. S. Schlau-Cohen, C. M. Isborn, and T. E. Markland. Two-dimensional electronic spectroscopy in the condensed phase using equivariant transformer accelerated molecular dynamics simulations, *J. Phys. Chem. Lett.*, (2025)
- 8 S. Chennakesavalu, F. **Hu**, S. Ibarra, and G. M. Rotskoff. Aligning Transformers with Continuous Feedback via Energy Rank Alignment. arXiv:2405.12961, *ICLR 2025 Workshop on Generative and Experimental Perspectives for Biomolecular Design* (2025)
- 9 F. **Hu**, M. S. Chen, G. M. Rotskoff, M. W. Kanan, and T. E. Markland. Accurate and efficient structure elucidation from routine one-dimensional NMR spectra using multitask machine learning. *ACS Cent. Sci.*, **10**, 11, 2162–2170 (2024)
- 10 S. Chennakesavalu, S. K. Manikandan, F. **Hu**, and G. M. Rotskoff. Adaptive nonequilibrium design of actin-based metamaterials: Fundamental and practical limits of control. *Proc. Natl. Acad. Sci.*, **121** (8), e2310238121 (2024)
- 11 P. Eastman, R. Galvelis, R. P. Peláez, C. R. A. Abreu, S. E. Farr, E. Gallicchio, A. Gorenko, M. M. Henry, F. **Hu**, J. Huang, A. Krämer, J. Michel, J. A. Mitchell, V. S. Pande, J. PGLM Rodrigues, J. Rodriguez-Guerra, A. C. Simmonett, S. Singh, J. Swails, P. Turner, Y. Wang, I. Zhang, J. D. Chodera, G. De Fabritiis, and T. E. Markland. OpenMM 8: Molecular Dynamics Simulation with Machine Learning Potentials. *J. Phys. Chem. B*, **128**, 1, 109-116 (2024)
- 12 F. **Hu**, F. He, and D. J. Yaron. Treating Semiempirical Hamiltonians as Flexible Machine Learning Models Yields Accurate and Interpretable Results. *J. Chem. Theory Comput.*, **19**, 18, 6185–6196 (2023)

Presentations

- 1 (Poster) Efficient, Few-Shot Directed Evolution With Energy Rank Alignment. International Conference on Learning Representations (ICLR) 2026, Rio de Janeiro, Brazil.
- 2 (Invited talk) Using AI to push the limits of 1D NMR for structure elucidation. Novartis Graduate Colloquium 2026, Novartis, Cambridge, MA
- 3 (Poster) Pushing the limits of one-dimensional NMR spectroscopy for automated structure elucidation using artificial intelligence. Berkeley Statistical Mechanics Meeting 2026, University of California, Berkeley, CA
- 4 (Poster) Aligning Transformers with Continuous Feedback via Energy Rank Alignment. Conference on Neural Information Processing Systems (NeurIPS) 2025, San Diego, CA
- 5 (Poster) Accurate and Efficient Structure Elucidation from NMR Using Artificial Intelligence. SMASH 2025, Porto, Portugal
- 6 (Contributed talk) Automated structure elucidation from NMR using artificial intelligence. Young WATOC (World Association of Theoretical and Computational Chemists) 2025, Oslo, Norway
- 7 (Poster) Energy rank alignment. International Conference on Learning Representations (ICLR) 2025, Singapore
- 8 (Poster) Accurate and efficient structure elucidation from routine one-dimensional NMR spectra using multitask machine learning. Experimental Nuclear Magnetic Resonance Conference 2025, Pacific Grove, CA
- 9 (Poster) Structure elucidation from NMR using machine learning. Berkeley Statistical Mechanics Meeting 2025, University of California, Berkeley, CA
- 10 (Poster) Accurate and efficient structure elucidation from routine one-dimensional NMR spectra using multitask machine learning. American Conference on Theoretical Chemistry 2024, University of North Carolina, Chapel Hill, NC
- 11 (Poster) Accurate and efficient structure elucidation from routine one-dimensional NMR spectra using multitask machine learning. West Coast Theory Meeting 2024, University of California, Merced, CA
- 12 (Contributed talk) Semiempirical Hamiltonians as Flexible Model Forms for Machine Learning. ACS 2023, San Francisco, CA
- 13 (Poster) DFTBML: A Machine-Learned Density Functional Based Tight Binding Model for the Deep Learning of Chemical Hamiltonians and Properties. West Coast Theory Meeting 2023, University of California, Davis, CA
- 14 (Contributed talk) DFTBML: A Machine-Learned Density Functional Based Tight Binding Model for the Deep Learning of Chemical Hamiltonians and Properties. Beckman Symposium 2022, Rutgers University, NJ

HONORS AND AWARDS

John Stauffer Memorial Fellowship <i>Stanford University, Stanford, CA</i>	2026
Center for Molecular Analysis and Design (CMAD) Fellowship <i>Stanford University, Stanford, CA</i>	2024-2026
Poster Prize <i>SMASH 2025, Porto, Portugal</i>	2025
Poster Prize <i>West Coast Theory Meeting 2024, University of California, Merced, CA</i>	2024
Poster Prize <i>West Coast Theory Meeting 2023, University of California, Davis, CA</i>	2023
NSF GRFP Honorable Mention <i>National Science Foundation</i>	2023
College Chemistry Award <i>Society of Analytical Chemists of Pittsburgh</i>	2022
Beckman Scholar <i>Arnold and Mabel Beckman Foundation</i>	2021-2022
National Merit Scholar <i>National Merit Scholarship Corporation</i>	2018-2022

LEADERSHIP AND SERVICE

Stanford Graduate Student Affairs Committee Industry Partnerships Co-Chair <i>Stanford University, Stanford, CA</i>	July 2025 - Present
◦ Networking and outreach with industry representatives to boost campus recruiting for chemistry PhD students	
Stanford FAST Mentor <i>Stanford University, Stanford, CA</i>	October 2024 - Present
◦ Scientific outreach work and mentoring underprivileged high school students doing science fair projects	
Teaching Assistant <i>Department of Chemistry, Stanford University, Stanford, CA</i>	August 2022 - May 2023, January 2025 - March 2025
◦ TA for 4 quarters of: general chemistry, statistical mechanics, quantum chemistry, and computational chemistry	
Head Teaching Assistant <i>School of Computer Science, Carnegie Mellon University, Pittsburgh, PA</i>	August 2021 - August 2022
◦ Head TA for an introductory computer science course	
◦ Developed course assignments and exams and arranged TA staff logistics in addition to teaching	